Rocket IEEE

Great question — and your decision depends on **your goal** and **resources**. Let's break it down clearly:

🚀 Should You Use Small Model Rocket Data or Large-Scale Rocket Data?

Goal	Recommendation
Publish a **realistic, innovative IEEE paper**	✅ **Yes — use small rockets**, and do it well
Want to work on **industry-scale (ISRO/NASA)** models	❌ Not feasible unless you have access to real large rocket datasets
Looking for a **physics + ML + education tool**	✅ Stay with model rocket scale
You want a **publishable project within a semester**	✅ Model rocket scale is better
You have **access to OpenRocket or test rig**	✅ Definitely stay with model rocket design

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🔍 Here's Why You Can Absolutely Stick with Model Rockets

✅ 1. Data Availability & Realism

- You already have usable data in grams and centimeters, which aligns with **model rocketry**.
- Data for **real large-scale rockets** is:
  - Hard to get (due to **security, cost, and proprietary** issues)
  - Very **complex** (multi-stage thrust, real-time telemetry, high-speed aerodynamics)

✅ 2. Control and Experimentation

- With model rockets, you can:
  - Simulate using **OpenRocket** or **RockSim**
  - Run **real field tests**
  - Modify and control designs easily

✅ 3. IEEE Encourages Scalable Educational Tools

IEEE loves:

- Low-cost educational innovations
- ML applied to simulations
- Design recommendation systems

> Your project could even be titled **"An ML-Based Predictive Framework for Stability in Low-Cost Model Rockets"** — which is very IEEE-sounding!

❌ Why Not Use Large Rocket Data?

- You'd need real telemetry, aerodynamic coefficients, stage separation data, etc.
- Without collaboration with **ISRO**, **NASA**, or **SpaceX**, it's nearly impossible to get real, labeled data.
- Even if you fake or simulate it, reviewers might **reject it as unrealistic**.

✅ What You Can Do Instead (to Make It Stronger)

If you stay with **small rocket data**, here's how to **make your IEEE paper impactful**:

🔧 Add Unique Angles:

- Train your model on **simulated flights (OpenRocket batch mode)** — hundreds of designs
- Add a **recommendation system**: suggest how to change CM or fin size for stability
- Compare your model with **existing physics rules** (like Barrowman equations)
- Deploy a **mini web tool** or GUI interface for designing rockets

📊 Validate It Properly:

- Compare your ML prediction vs. OpenRocket result or manual calculation
- Include **wind angle, launch pad effects**, etc., to make it more realistic

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✅ Final Recommendation

> **Stick with small model rocket data** — but go deep, make it smart, add automation or simulation, and connect it to real-world applications (education, field testing, design aid).

This is **practical**, **publishable**, and **innovative** — without overpromising.

Would you like help with:

- A proposed IEEE **paper title and abstract**?
- A **dataset expansion plan** using OpenRocket?
- A **workflow chart or diagram** for your paper?

I'm happy to help build this to IEEE level with you.